

Spam Mail Detector using ML.

① Import Libraries → Pandas
Numpy
Matplotlib
Seaborn

For NLP Preprocessing → nltk
string

stopwords, word-tokenize.

ML → train-test-split

Feature-extraction-text, Tfidf-
Vectorizer.

(naive-bayes → Multinomial NB

linear-model → Logistic Regression

metrics → Accuracy, Precision,
f1-score, confusion-matrix,
classification-report)

② Download NLP resources ('Punkt', 'stopwords')

③ Download the Dataset SMS Spam Collection &
Load the Dataset

④ Explore the Data Set

⑤ Convert Labels into Numbers (One Hot
Encoding)

⑥ Text Processing, Convert text into
↳ Convert text into
Lower case
↳ Remove Punctuation
↳ Tokenize
↳ Remove Stop Words

⑦ Convert Text into Numeric Features;
TF-IDF gives higher importance to meaningful words & lower importance to common words.

⑧ Split the Data into train-test

⑨ Train Naive Bayes (Multinomial NB)

⑩ Evaluate Naive Bayes Model Result,
Based on $\rightarrow$ Accuracy, Precision, F_1 -Score

⑪ Train Logistic Regression

⑫ Evaluate Logistic Regression Results,
based on $\rightarrow$ Accuracy, Precision,
 F_1 -Score,

⑬ Compare Both Models & Select the Best Model.

⑭ Report Writing.